

AI and Large Language Models Research Paper

Sofia Quaglia, Moisés A. Silva, Ciarán Neville

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AI and Large Language Model Research Paper (POL42560) – Generative Artificial Intelligence Statement

Tool	Purpose	Topic	Prompt
Grammarly (Word extension)	Grammar and spelling checks	Overall	None
ChatGPT (free version)	Formatting Overleaf Table - Table 1: Topics in Topic Modelling	Text analysis	Generate a LaTeX table in Overleaf that presents the results of a topic modeling analysis, including columns for the topic label, a brief topic description, top associated words, and the coherence score for each topic.
ChatGPT (free version)	Time stamps transcription	Text analysis	Write a Python script that extracts timestamped transcription segments from a Whisper model output and saves them to a CSV file using pandas. The output should include the start time, end time, and transcribed text for each segment.
ChatGPT (free version)	Set up Diarization and Transcription Chunk	Text analysis	Write a Python script that sets up the environment for audio diarization and transcription using Hugging Face pipelines. The script should include directory creation, model loading for speaker diarization (pyannote) and transcription (Whisper), GPU usage, and authentication via a Hugging Face token.
ChatGPT (free version)	Process One Chunk at the Time	Text analysis	Write a Python script that processes an individual audio chunk by performing speaker diarization and automatic speech transcription using preloaded models. The script should align speaker labels with timestamped transcription segments, construct a pandas DataFrame with start time, end time, text, and speaker columns, and save the output as a CSV file. Make sure to adjust chunk indexing and support repeated use across multiple chunks.
Elicit	Literature search	Overall	How do specific emotional display characteristics (positive vs negative facial expressions, vocal pitch variations) correlate with voter perception and electoral success in national-level political campaigns?
Perplexity	Draft refinement	Emotions in party politics	Review this draft on the role of emotions in party politics. Suggest improvements to how the ideas relate between them and do a grammar check.
Claude	Py-feat implementation	Facial expression analysis	I want to analyze an hour and a half long debate using pyfeat functionalities to identify facial activation units and the emotion associated with the combination of them. To help the models with identification I have two jpg images of both Trump and Harris if needed. Goal: Create an exhaustive step by step code intensive guide to create a notebook that fulfills this goal for an Apple Silicon setup, draw inspiration from these sources:

			Expressions from video: https://py-feat.org/basic_tutorials/02_detector_vids.html Full analysis: https://py-feat.org/basic_tutorials/04_fex_analysis.html I want to plot the facial activation units per candidate, and the evolution of emotions over the debate (lineplot with emotion and minutes), feel free to add appropriate additional visualizations Create a conda virtual environment for this to avoid dependencies issues and implement measures for optimization, I want the process to be as fast as possible for a whole 1.5 hour debate
ChatGPT 4.0	Diarization	Voice analysis	I have a folder of .wav files and I want Python code that runs speaker diarization on each one using pyannote.audio, skips any that are already processed, saves the results as .pkl files in another folder, uses the GPU, and shows progress.
ChatGPT 4.0	Speaker Validation	Voice analysis	I have a full debate audio file and a saved diarization result with speaker names. I want to extract a few short audio clips for certain speakers so I can listen and check if the speaker labels are accurate. The code should save the clips locally and also copy them to a folder in my Google Drive
ChatGPT 4.0	Manual speaker validation	Voice analysis	I have diarization results from a debate and the full debate audio file. I want to go through any speakers who aren't labeled as Donald Trump, Kamala Harris, or the Moderator, and export a few short audio clips for each of them so I can check who they are. The clips should be saved in a folder in my Google Drive for manual review.
ChatGPT 4.0	Emotion recognition	Voice analysis	I have a full debate audio file and speaker diarization results with speaker IDs. I also have a mapping of those IDs to real names. I want to go through each segment that belongs to Trump or Harris, run emotion recognition using a pre-trained model from Hugging Face (wav2vec2 for emotion), and save the start time, end time, speaker name, and predicted emotion into a CSV file in my Google Drive
ChatGPT 4.0	Transcription from audio	Voice analysis	I want to transcribe a debate audio file using the faster-whisper model. The code should automatically check if a GPU is available and use it if possible, then load the large-v2 model, transcribe the audio with voice activity detection, and save the full transcript with timestamps to a text file
ChatGPT 4.0	BERTopic	Voice analysis	I have a transcript of a debate with timestamps, and I want to use BERTopic to identify the main topics discussed. The code should load the transcript, split it into timestamped segments, run BERTopic on each segment, save a CSV with the timestamp, text, topic ID, and topic name, and also save the trained topic model to my Google Drive

ChatGPT 4.0	Coherence score	Voice analysis	Coherence scores: I used BERTopic to find topics in my debate transcript and now I want to evaluate the quality of each topic. The code should skip certain noisy topics, build word lists for the rest, compute coherence scores using Gensim, merge those scores with topic names, and then save the top 10 most coherent topics to a CSV file in my Google Drive
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